When a Small Krylov Space Can Solve a Nonconvex Model

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Analysis of Krylov Subspace Solutions of Regularized Non-Convex Quadratic Problems

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Survey: Escape methods

Spreadsheet priority: None; independent recommendation: Skim

The paper in one minute

Question: How quickly can matrix-vector products solve trust-region and cubic-regularized quadratic subproblems?

Model/Method: Globally minimize the model over span$\{b,Ab,...,A^{t-1}b\}$, computed efficiently by Lanczos; randomize to uncover hidden negative curvature.

Finding: Theorems give exponential error decay with shifted conditioning and a sublinear t^{-2} bound with an alignment logarithm. Constructed lower bounds show the rates are sharp in their oracle setting.

Why we care: The inner cubic solve is a major part of second-order optimization cost. A gradient-seeded Krylov method can miss a negative eigenspace completely in the hard case.

1. Start from the model that must be solved

$$q(x) = (1/2)x^T Ax + b^T x,\ A = A^T.$$

The trust-region problem minimizes q under $\|x\| \le R$. Cubic regularization instead minimizes $q(x) + (\rho/3)\|x\|^3$. A can be indefinite, but the radius or cubic term prevents unbounded descent. In an outer Newton-type method, b is a local gradient and A a local Hessian. Solving this model globally is not the same as globally minimizing the original nonlinear training loss.

A huge dense A may be impossible to store, yet Av may be affordable. A Krylov method constructs $K_t(A,b) = \text{span}\{b,Ab,...,A^{t-1}b\}$, then solves the model over this small space. Lanczos builds an orthonormal basis Q_t whose projected matrix $Q_t^T A Q_t$ is tridiagonal. The expensive operations are roughly t products with A; the small projected model must still be solved.

2. Why polynomials appear

Every vector in K_t is p(A)b for a polynomial p of degree at most t-1. In an eigenbasis of A, its ith coordinate is $p(\lambda_i)b_i$. Thus a polynomial can rescale a component that exists in b, but it cannot create an eigenvector component whose coefficient b_i is exactly zero. This simple fact explains both rapid approximation and the hard-case failure.

Fixed matrix, adaptive directions. All vectors in one Krylov sequence are built using the same A. Replacing A by a freshly sampled Hessian on every product changes the method and is not covered by these subspace bounds.

3. Worked example: a hidden negative eigendirection

Pedagogical illustration, constructed for these notes.

Let A = diag(-2,1), b = (0,1), and R = 1. Then $q(x,y) = -x^2 + y^2/2 + y$. The first coordinate has negative curvature, but b has no component there.

$A^j b = b$ for every $j \geq 0$, so $K_t(A,b) = \text{span}\{(0,1)\}$ for every t. Restricting x = 0 gives $q(0,y) = y^2/2+y$, minimized on $|y| \leq 1$ at y = -1, with value -1/2. More products cannot improve this space.

For a full-space minimum choose the trust-region multiplier $\lambda^* = 2$. The conditions $(A+\lambda^* I)s^* = -b$ and $\|s^*\| = 1$ give $3y = -1$ and $x^2 = 1-1/9 = 8/9$.

$$s^* = (\pm 2\sqrt{2}/3, -1/3),\ q(s^*) = -8/9 + 1/18 - 1/3 = -7/6.$$

The restricted gap is (-1/2)-(-7/6) = 2/3 forever. The shifted matrix diag(0,3) is positive semidefinite, so the stated global optimality conditions hold. A local projected solve is correct within its space yet globally wrong for the original model.

If an extra vector v has a nonzero first coordinate, span{b,v} is all of R^2. A continuously sampled random direction has this property with probability one, so the joint space can represent s*. This small example demonstrates why adding a random direction repairs the missing-component problem.

The hard case is more specific than a poorly conditioned positive-definite problem: $u^T b = 0$ and $\lambda^* = -\lambda_{\min}(A)$. Here both hold exactly. Near the hard case, a tiny but nonzero component is available; the nonconvex sublinear bound tracks the difficulty through an alignment logarithm rather than requiring a uniformly large initial projection.

4. The two convergence bounds

Theorem1 bounds the objective gap of the exact minimizer over K_t for the trust-region problem. Write $\Delta_0 = q(0)-q(s^*)$ and $\kappa = (\lambda max+\lambda^*)/(\lambda min+\lambda^*)$ when the denominator is positive. One bound is

$$q(s_t)-q(s^*) \leq 36\Delta_0 \exp(-4t/\sqrt{\kappa}).$$

This is geometric error reduction in the subspace order, with a condition number for the shifted problem, not for A alone. A second bound is proportional to $(\lambda max-\lambda min)\|s^*\|^2/t^2$, multiplied in the nonconvex case by a logarithmic factor in $\|b\|/|u_{min}^T b|$. Both deterministic bounds can be vacuous in the exact hard case, as the worked example confirms.

5. What makes the proof work

Krylov spaces are invariant under scalar shifts: $K_t(A,b) = K_t(A+\lambda I,b)$. Shifting to a positive-semidefinite matrix permits convex approximation estimates. The gap decomposes into a shifted convex-model error plus a norm discrepancy. Polynomial inverse approximation controls the first term; a good extremal-eigenvector approximation or a small rescaling controls the second. Chebyshev polynomial bounds explain the $\exp(-4t/\sqrt{\kappa})$ and t^{-2} scales.

6. Randomization has a cost and a choice

Corollary2 augments the space with a random direction and its matrix powers. With probability at least 1-δ, the alignment difficulty becomes a logarithmic factor in dimension and failure probability. Counting both sequences matters: about half the products are spent on the extra vector. Corollary3 instead perturbs b by σv. It avoids that doubling but adds an objective-error term 2σR. Choosing σ too large imposes a high error floor; too small can delay discovery of negative curvature.

Theorem5 constructs difficult instances whose bounds match the stated scales up to constants. A resisting-oracle argument extends appropriate lower bounds to deterministic matrix-vector methods with sufficient dimension. This is not a lower bound against all randomized methods; randomization explicitly helps the nonconvex alignment obstruction.

7. Transfer to cubic regularization

For the cubic convention $\rho\|x\|^3/3$, global optimality requires $(A+\rho\|s^*\|I)s^* = -b$ and $A+\rho\|s^*\|I$ positive semidefinite. The equivalent trust-region instance has $R = \|s^*\|$ and $\lambda^* = \rho\|s^*\|$. At feasible points the cubic objective gap is no larger than the corresponding trust-region gap; Corollary4 transfers the convergence analysis. This equivalence is analytical: the algorithm is not assumed to know $\|s^*\|$ beforehand.

8. What was tested

Section5 generates 5,000 random diagonal cubic problems of dimension one million for each of three prescribed finite condition numbers. Rotation invariance makes diagonal instances useful for testing subspace behavior while keeping products cheap. Figure1 supports the two regimes in these ensembles. Additional hard-case experiments compare joint spaces with perturbing b; fixed perturbations reach an error floor while joint spaces can continue improving.

SupplementE specifies uniform interior eigenvalues, controlled extrema and scaling to unit initial gap. Hard-case instances deliberately set $b_1 = 0$; the supplementary figure varies eigen-gap. These are generated quadratic-model experiments, not neural-network training comparisons. Claims about outer ARC complexity on main page8 are expressly discussed without a full outer analysis in this paper.

9. Limits and survey use

The theory analyzes exact minimizers of projected problems and mathematical Krylov spaces. Finite precision can destroy orthogonality; exact d-step termination is not a reliable floating-point promise. Inner reduced solves, storage and possible basis regeneration also cost work. The randomization bounds concern a fixed symmetric matrix, not arbitrary changing stochastic products. This paper supplies an inner-solver ingredient, not an unconditional end-to-end saddle-escape theorem for neural networks.

10. Quick reference

Agarwal2017 also studies efficient cubic solves, through shifted systems and eigenvectors. Here their ingredients already lie in an appropriate Krylov space. Tripuraneni2018 additionally controls stochastic derivative errors and outer stopping; substituting this inner method would require proving that its particular approximation contract is met.

Term	Meaning
κ	Conditioning after the optimal regularization shift.
Hard case	Missing minimum-eigenvector component and singular optimal shift.
Random b perturbation	Improves alignment but introduces controllable bias.

Source: main PDF1-9, Theorem1, Corollaries2-4, Theorem5; all12 supplement pages (printed12-23). Trust-region example constructed and checked independently.

Papers Master - Our notes (copy): Krylov trust-region/cubic subproblem solves have geometric and t^-2 objective-gap bounds controlled by shifted conditioning and eigenvector alignment. The exact hard case can stall forever; random augmentation or perturbation repairs alignment, with extra products or bias. Results concern fixed-matrix inner models, not global neural-network training.

Reading recommendation: Study the Krylov mechanism, hard-case example and randomization. Keep this inner-subproblem result separate from an outer neural-network convergence theorem.